Algebraic properties of p-adic numbers

In this section we fix a prime number p.

We have introduced the p-adic absolute value $|\cdot|_p$ on the field $\mathbb{Q}$ of rational numbers. We have also seen that the completion $\mathbb{Q}_p$ with respect to this absolute value gives a field which we call the field of p-adic numbers. In this section we explore the algebraic properties of this field.

p-adic Valuation

Given a non-zero rational number $r \in \mathbb{Q}^*$, we write it in the form $r = p^k(m/n)$ where k, m, n are integers, $n > 0$ and $\gcd(p, m) = 1 = \gcd(p, n)$. In that case $|r|_p = p^{-k}$. This depends *only* on k. Thus, it is natural to define $v_p(r) = k$, when r has the above form. This function $v_p : \mathbb{Q}^* \to \mathbb{Z}$ can be extended by defining $v_p(0) = \infty$ by expanding the range to be $\mathbb{Z} \cup \{\infty\}$. This function $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ is called the p-adic valuation. We note that if we *define* $p^{-\infty} = 0$, then we have the identity

$$|r|_p = p^{-v_p(r)}$$

for all r in $\mathbb{Q}$.

Note that if $r = p^k(m/1)$ is an *integer* then $v_p(r) = k \geq 0$. Moreover, $v_p(r \cdot s) = v_p(r) + v_p(s)$ for all rational numbers r and s providing we define $a + \infty = \infty$ for any element a of $\mathbb{Z} \cup \{\infty\}$. Moreover, note that $v_p(m)$ of an integer m is 0 if and only if it is co-prime to p.

If r, s are two non-zero rational numbers with $v_p(r) = k$ and $v_p(s) = t$, then $r = p^k(m/n)$ and $s = p^t(u/v)$ where m, n, u, v are integers co-prime to p with $n > 0$ and $v > 0$. Without loss of generality, let us assume that $k \leq t$. Then we have

$$r + s = p^k \frac{m \cdot v + p^{t-k} n \cdot u}{n \cdot v}$$

Since $\gcd(p, n \cdot v) = 1$, we see that $v_p(v \cdot v) = 0$. Hence,

$$v_p(r + s) = k + v_p(m \cdot v + p^{t-k} n \cdot u) \geq k = \min\{k, t\} = \min\{v_p(r), v_p(s)\}$$

We thus obtain $v_p(r+s) \geq \min\{v_p(r), v_p(s)\}$. This inequality can also be checked when r or s is 0 by declaring ∞ as the largest element of $\mathbb{Z} \cup \{\infty\}$.

Further, we note that if $t > k$, then $\gcd(p, m \cdot v + p^{t-k} n \cdot u) = \gcd(p, m \cdot v) = 1$. So, in this case we have $v_p(r + s) = k = v_p(r)$. In other words, we have proved:

$v_p(r + s) \geq \min\{v_p(r), v_p(s)\}$ and equality holds if $v_p(r) \neq v_p(s)$.

Since $|r|_p = p^{-v_p(r)}$, this statement can be re-written as

$|r + s|_p \leq \max\{|r|_p, |s|_p\}$ and equality holds if $|r|_p \neq |s|_p$.

It is not difficult to show that $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ is the *unique* function satisfying the following properties:

1. $v_p(r) = \infty$ if and only if $r = 0$.
2. $v_p(r \cdot s) = v_p(r) + v_p(s)$.
3. $v_p(p) = 1$.
4. $v_p(n) = 0$ if n is an integer such that $\gcd(p, n) = 1$.

Cauchy sequences

A Cauchy sequence (r_n) with respect to $|\cdot|_p$ is one that satisfies the following property:

> For all integers k, there is an integer N such that, for all $m, n \geq N$, we have $|r_n - r_m|_p < 1/k$.

Since $|\cdot|_p$ takes values in $\{p^k : k \in \mathbb{N}\} \cup \{0\}$, we can alternatively write:

> For all integers k, there is an integer N such that, for all $m, n \geq N$, we have $|r_n - r_m|_p \leq p^k$.

In terms of the p-adic valuation, this can be written as:

> For all integers k, there is an integer N such that, for all $m, n \geq N$, we have $v_p(r_n - r_m) \geq k$.

Note that the inequality is reversed due to the minus sign in the identity $|r|_p = p^{-v_p(k)}$.

Since $|\cdot|_p$ is a continuous function on $\mathbb{Q}$, it extends to a continuous function on $\mathbb{Q}_p$. Thus, if (r_n) converges to $x \in \mathbb{Q}_p$, then the sequence $(|r_n|_p)$ of *real* numbers converges to the real number $|x|_p$.

We now separate this into two cases.

Case $|x|_p = 0$. In this case, we see that $(|r_n|_p)$ converges to 0. This means that:

> For all integers k, there is an integer N such that, for all $n \geq N$, we have $v_p(r_n) \geq k$.

It follows that (r_n) converges to 0. Hence $x = 0$.

Note that the above statement is the same as saying that $v_p(r_n)$ goes to infinity as n goes to infinity.

Case $|x|_p \neq 0$. In this case $(|r_n|_p)$ converges to a positive number $|x|_p = \beta > 0$.

Since p^{-k} converges to 0 as k goes to infinity, it follows that there is a k such that $\beta > p^{-k}$.

Thus, there is an N such that, for all $n \geq N$, we have $|r_n|_p > p^{-k}$. This means that, for $n \geq N$, we have $v_p(r_n) < k$.

Since (r_n) is a Cauchy sequence, there is an M such that, for all $m, n \geq N$, we have $v_p(r_m - r_n) > k$.

Thus, for all $m, n \geq N_0 = \max M, N$, we see that $v_p(r_m - r_n) > k$ and $v_p(r_n) < k$. Since $r_m = r_n + (r_m - r_n)$, and $v_p(r_n) < v_p(r_m - r_n)$, we see that $v_p(r_m) = v_p(r_n)$. Hence, for all $m, n \geq N_0$, we see that $v_p(r_m) = v_p(r_n)$ is a *constant* a.

Since $|r_n|_p = p^{-a}$ for all $n \geq N_0$, we see that $|x|_p = p^{-a}$ by continuity. We thus extend v_p as a function $v_p : \mathbb{Q}_p \to \mathbb{Z} \cup \{\infty\}$ by defining $v_p(x) = a$ in this case.

In summary, we see that if (r_n) is a Cauchy sequence with respect to $|\cdot|_p$, then one of the following statements holds:

a. $v_p(r_n)$ goes to infinity as n goes to infinity.
b. $v_p(r_n)$ is constant when n is sufficiently large.

A different way to say the same thing is to introduce a topology on $\mathbb{Z} \cup \{\infty\}$ with ∞ as the limit point of all increasing sequences of integers. Then, v_p as defined above is a *continuous* function $\mathbb{Q}_p \to \mathbb{Z} \cup \{\infty\}$.

By similar arguments, we shee that the following holds for p-adic numbers x and y:

$$v_p(x + y) \geq \min\{v_p(x), v_p(y)\} \text{ and equality holds if } v_p(x) \neq v_p(y).$$

$\mathbb{Z}_{(p)}$ and $\mathbb{Z}_p$

For k any integer, we see that the set $I_k = \{r : v_p(r) \geq k\}$ is closed under addition. Moreover, we see that $I_k = p^k I_0$.

Since $v_p(r \cdot s) = v_p(r) + v_p(s)$ we see that I_0 is closed under multiplication as well. Since 0 and 1 lie in I_0, we see that it is a ring. We use the notation $\mathbb{Z}_{(p)}$ for this ring. (This notation comes from the notion of *localisation* in commutative algebra.)

Since $|r|_p = p^{-v_p(r)}$, we see that $I_k = \{r : |r|_p \leq p^{-k}\}$.

Since $|\cdot|_p$ is a *continuous* function on $\mathbb{Q}$ and $\mathbb{Q}_p$, we that the closure of I_k is

$$\overline{I_k} = \{x \in \mathbb{Q}_p : |x|_p \leq p^{-k}\}$$

Since multiplication by p^k is continuous for all k, we see that $\overline{I_k} = p^k \overline{I_0}$.

It also follows from the above inequality that these are subgroups of $\mathbb{Q}_p$.

Note that $\overline{I_0}$ consists of $x \in \mathbb{Q}_p$ such that $|x|_p \leq 1$ and $\overline{I_1}$ consists of $x \in \mathbb{Q}_p$ such that $|x|_p \leq (1/p) < 1$.

It is clear that $\overline{I_0}$ is *also* closed under multiplication and contains 0 and 1. Thus, it is a ring. We use the notation $\mathbb{Z}_p$ for $\overline{I_0}$ and call it the ring of p-adic integers.

Note that what we have so far been calling I_k is $p^k \mathbb{Z}_{(p)}$ and what we have called $\overline{I_k}$ is $p^k \mathbb{Z}_p$.

p-adic expansion

Given a rational number $r = p^k(m/n)$ where k, m and $n > 0$ are integers with m and n co-prime to p. Using the GCD algorithm, we can find a and b integers such that $n \cdot a + p \cdot b = 1$. We then calculate

$$p^k \frac{m}{n} - p^k(m \cdot a) = p^k m \frac{1 - n \cdot a}{n} = p^{k+1} m \frac{b}{(} n)$$

It follows that r lies in $p^k(m \cdot a) + p^{k+1}\mathbb{Z}_{(p)}$ inside $p^k\mathbb{Z}_{(p)}$. Note that $(m \cdot a)$ is an integer, so we can write it as $c + p \cdot d$ where $c \in \{0, 1, \ldots, p-1\}$. Substituting this expression for a we see that:

> Given an element r in $p^k\mathbb{Z}_{(p)}$, there is a $c \in \{0, 1, \ldots, p-1\}$ such that r lies in $p^k c + p^{k+1}\mathbb{Z}_{(p)}$.

Moreover, if c and c' are *distinct* elements of $\{0, 1, \ldots, p-1\}$, then $c - c'$ is not divisible by p so that $p^k c + p^{k+1}\mathbb{Z}_{(p)}$ and $p^k c' + p^{k+1}\mathbb{Z}_{(p)}$ are *disjoint*. Thus, we see that $p^k\mathbb{Z}_{(p)}$ is the disjoint union of $p^k c + p^{k+1}\mathbb{Z}_{(p)}$.

We can say this better in terms of group theory. The quotient of the group $p^k\mathbb{Z}_{(p)}$ by the subgroup $p^{k+1}\mathbb{Z}_{(p)}$ is *isomorphic* to the group $\mathbb{Z}/\langle p \rangle$, where the isomorphism sends c in $\mathbb{Z}/\langle p \rangle$ to the coset $p^k c + p^{k+1}\mathbb{Z}_{(p)}$.

Given a p-adic number x, we have a Cauchy sequence (r_n) of rational numbers that converges to x. Thus, if $v_p(x) = k$, there is an N such that $v_p(x - r_n) \geq k+1$ for all $n \geq N$. Note that this means that $v_p(r_N) = k$. Thus, we can find $c_k \in \{0, 1, \ldots, p-1\}$ so that r_N lies in $p^k c_k + p^{k+1}\mathbb{Z}_{(p)}$. We easily check that $v_p(x - p^k c_k) \geq k + 1$.

Denote $x_k = x$ and $x_{k+1} = x - p^k c_k$ with $c_k \in \{0, 1, \ldots, p-1\}$. Iterating this process, we produce a sequence $(c_j)_{j \geq k}$ of elements of $\{0, 1, \ldots, p-1\}$ such that $v_p(x - (p^k c_k + p^{k+1} c_{k+1} + \cdots + p^m c_m) \geq m + 1$. If we define $s_m = p^k c_k + p^{k+1} c_{k+1} + \cdots + p^m c_m$, then we see that (s_m) is a sequence of rational numbers (k could be negative!) which converge to x in $|\cdot|_p$.

Conversely, suppose we are given a sequence $(c_j)_{j \geq k}$ of elements of $\{0, 1, \ldots, p-1\}$. If we define $s_m = p^k c_k + p^{k+1} c_{k+1} + \cdots + p^m c_m$, then we see that $v_p(s_m - s_n) \geq \min\{n+1, m+1\}$. So (s_m) is a Cauchy sequence of rational numbers in the p-adic norm.

> Every p-adic number can be written uniquely in the form $\sum_{j=k}^{\infty} p^j c_j$ with c_j in $\{0, 1, \ldots, p-1\}$ for each j.

p-adic arithmetic

Given any element c of $\mathbb{Z}/\langle p^{n+1} \rangle$ it can be uniquely expressed in the form $c_0 + pc_1 + \cdots + p^n c_N$ where c_j's are elements of $\{0, 1, \ldots, p-1\}$.

In particular, if $c = a + b$, where a and b are elements of $\mathbb{Z}/\langle p^{n+1}\rangle$ which are written as $a = a_0 + pa_1 + \cdots + p^n a_n$ and $b = b_0 + pb_1 + \cdots + p^n b_n$, then we can find such an expression for c. This can be made algorithmic as follows:

1. We define $q_0 = 0$ and set $j = 0$.
2. We write $a_j + b_j + q_j = q_{j+1}p + c_j$ where $0 \le c_j < p$.
3. If $j < n$ then we replace j by $j + 1$ and go back to step 1.

Note that the value of c_j *only* depends on the value of $a_0, \ldots, a_j$ and $b_0, \ldots, b_j$. The "carry over" q_{j+1} only affects c_k for $k \ge j + 1$.

We can apply this to p-adic numbers as follows. Given any p-adic number z, we can use the previous section to write it in the form $z = (p^k c_k + p^{k+1} c_{k+1} + \cdots + p^m c_m) + z_{m+1}$ where z_{m+1} lies in $p^{m+1}\mathbb{Z}_p$. As seen above, this expression is unique.

Given x and y two p-adic numbers written as

$$x = (p^k a_k + p^{k+1} a_{k+1} + \cdots + p^m a_m) + x_{m+1}$$
$$y = (p^k b_k + p^{k+1} b_{k+1} + \cdots + p^m b_m) + y_{m+1}$$

where x_{m+1} and y_{m+1} are in $p^{m+1}\mathbb{Z}_p$ and a_i's and b_i's are integers. Using the algorithm above, we find c_j's (after renumbering the subscripts by subtracting k) and check that

$$x + y = (p^k c_k + p^{k+1} c_{k+1} + \cdots + p^m c_m) + p^{m+1} q_{m+1} + x_{m+1} + y_{m+1}$$

Thus, if $z = x + y$, then $z_{m+1} = p^{m+1} q_{m+1} + x_{m+1} + y_{m+1}$.

It is worth noting that c_j *only* depends on $a_k, \ldots, a_j$ and $b_k, \ldots, b_j$. In particular, the "carry over" does not affect the correctness of the p-adic expansion. This is *different* from the situation for real numbers where the "carry over" from "later" decimal places can affect the correctness of "earlier" decimal places. In other words, p-adic arithmetic is algorithmically *exact* whereas errors accumulate in algorithmic real arithmetic which can only work with finite decimal expansions.

The product of $x = \sum_{j=m}^{\infty} p^j a_j$ and $y = \sum_{j=n}^{\infty} p^j b_j$ can first be written as $z = x \cdot y = \sum_{j=m+n}^{\infty} p^j d_j$, where

$$d_j = \sum_{i=m}^{j-i} a_i b_{j-i}$$

(following the convention that $b_t = 0$ if $t < n$). Note that this is a *finite* sum and so d_j's are integers. We then *reduce* this to the required form $z = (p^{m+n} c_{m+n} + p^{m+n+1} c_{m+n+1} + \cdots + p^t c_t) + z_{t+1}$ by the following algorithm:

1. Put $j = m + n$ and $q_j = 0$.
2. Write $d_j = q_{j+1}p + c_j$ where $0 \le c_j < p$.
3. If $j < t$, then replace j by $j + 1$ and go to step 1.

One again, we note that that we need only finitely many a_j's and b_j's to *precisely* determine c_j's upto $j \le t$.